

FastAPI Dependency Version Conflict SOP

Objective

Recover FastAPI applications failing due to incompatible Python libraries, transitive dependency conflicts, or runtime version mismatch.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
3. Permission to view application configuration, environment variables, and deployment status.
4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T037
Issue	FastAPI dependency version conflict
Category	Deployment
Service	FastAPI
Severity	Medium
Status	Resolved
Description	App fails due to incompatible libraries
Expected Resolution	Update dependency versions

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, API gateway logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Confirm failure started after dependency, image, or environment change.
2. Review logs for import errors, incompatible versions, or missing symbols.
3. Compare current dependency lock file with last working version.
4. Check FastAPI, Starlette, Pydantic, Uvicorn, and Python version compatibility.
5. Reproduce installation in a clean environment or CI job.
6. Pin compatible versions or upgrade dependent libraries together.
7. Rebuild and redeploy the application image.
8. Run startup, health, and endpoint smoke tests before closing ticket.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
```

```
kubectl get pods -n <namespace> | grep fastapi
kubectl logs <fastapi-pod> -n <namespace> --tail=100
kubectl describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Import error	Library version conflict	Pin compatible dependency versions
App fails after upgrade	Breaking change in dependency	Review release notes and adapt code
Local works, server fails	Environment mismatch	Align Python and package versions
Pydantic errors	Major version incompatibility	Update models or pin version

Best Practices

1. Use dependency lock files.
2. Run dependency checks in CI.
3. Upgrade framework libraries deliberately.
4. Keep rollback-ready image tags.

References

FastAPI Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | API Gateway Logs | Backend Runbook